

EKF-Adaptive False-Alarm-Aware Sensor Placement with a CBF Safety Filter for Barrier Coverage

Research Task Report 2 — Response to Prof. Ben Kim

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1. Problem Background and Motivation

This report builds directly on two papers assigned by Prof. Kim:

- **Paper [1]:** Kim et al., “Near-Optimal Sensor Placement for Detecting Stochastic Target Trajectories in Barrier Coverage Systems,” IEEE SysCon 2025. This paper models target trajectories as a Log-Gaussian Cox Line Process (LGCLP) and deploys K sensors to maximize the void probability—the probability that every target crossing the barrier is detected by at least one sensor. A greedy algorithm on a Jensen lower bound achieves at least $\left(1 - \frac{1}{e}\right)$ of optimal.
- **Paper [2]:** Kim et al., “Robust Sensor Placement for Poisson Arrivals with False-Alarm-Aware Spatiotemporal Sensing,” arXiv:2510.05343. This paper generalizes the framework to spatio-temporal Poisson arrivals. Each sensor i is equipped with a tunable filter parameter θ_i (e.g., a Kalman or particle-filter threshold/gain) that improves raw detection probability p but simultaneously inflates the false-alarm rate through an availability function $\alpha(\chi) = \frac{1}{1+\beta\chi}$. The effective detection probability is $\tilde{p}_i = \alpha(\chi(\theta_i, \omega)) \cdot p_i$, where $\omega(s, t) \in [0, 1]$ is a stochastic spatio-temporal environmental loss field.

The key limitation of Paper [2]: the filter gain θ_i is chosen offline, once, using the prior distribution of $\omega(s, t)$, and is never updated as the true environment evolves. In real deployments—turbidity gradients shifting with currents, weather fronts drifting across a surveillance corridor—the optimal θ drifts with the environment. A static θ permanently pays a suboptimal false-alarm-versus-detection trade-off that an online estimator can avoid.

This is the gap addressed here, and it is exactly where an Extended Kalman Filter (EKF) for online ω -estimation combined with a Control Barrier Function (CBF) safety filter on θ naturally fits.

2. Summary of the Two Reference Papers

2.1 Paper [1] — Barrier Coverage via LGCLP (SysCon 2025)

Target trajectories are modelled as a Log-Gaussian Cox Line Process (LGCLP). Every linear track $y = mx + b$ in inertial space is bijectively mapped to a point (α, p) in a 2-D representation space C via:

$$T: y = mx + b \rightarrow \left(\alpha = \frac{\pi}{2} + \arctan(m), p = \frac{b}{\sqrt{1 + m^2}} \right).$$

Under this mapping the line process becomes a point process in C , and sensor detection probabilities transform accordingly. The optimization objective is the void probability:

$$v(a) = E\lambda \left[\exp \left(- \int_{\{T_C\}} \lambda(l) \pi_{C(l,a)} dl \right) \right]$$

Because maximizing v exactly is NP-hard, the method maximizes a Jensen lower bound greedily, achieving at least $\left(1 - \frac{1}{e}\right)$ of the optimal. The unknown intensity function $\lambda(l)$ is estimated from AIS data using Integrated Nested Laplace Approximation (INLA).

2.2 Paper [2] — False-Alarm-Aware Spatiotemporal Sensing (arXiv:2510.05343)

Paper [2] generalizes the void-probability framework to spatio-temporal Poisson arrivals with a stochastic environmental field $\omega(s, t)$ drawn from a Gaussian process. The key contributions are:

- A unified availability model $\alpha(\chi) \cdot p$ coupling detection with false-alarm cost.
- A pointwise sufficient condition (Theorem 1) for when increasing the filter gain θ is beneficial:

$$\frac{\partial_{\theta} p}{p} \geq - \left(\frac{\alpha'(\chi)}{\alpha(\chi)} \right) \partial_{\theta} \chi.$$
- A new coverage-oriented lower bound on void probability that strictly outperforms the classical $\left(1 - \frac{1}{e}\right)$ guarantee in a low-coverage regime.
- Robustness guarantees when intensity and ω are estimated from finite samples (Theorems 2–3).

Crucially, the filter gain θ_i is chosen offline and held static—selected once knowing the prior distribution of $\omega(s, t)$, and never updated as the environment actually evolves. This motivates the extension proposed here.

3. Proposed Extension — EKF-Adaptive θ with CBF Safety

We make the filter gain $\theta_i(t)$ online and EKF-driven, and we use a CBF safety filter to guarantee that this online adaptation never lets the instantaneous void probability fall below a user-specified threshold v_{min} the three components are:

3.1 EKF on ω at Each Sensor

Each sensor i maintains a scalar EKF whose state is its local environmental loss $\hat{\omega}_i(t)$. Measurements y_i, k come from the sensor's own SNR or hit-rate statistics (in simulation, we feed noisy ground truth). The state-space model per sensor is:

$$\begin{aligned} x_i, k+1 &= F \cdot x_i, k + w_k, \quad w_k \sim N(0, Q) \\ y_i, k &= x_i, k + v_k, \quad v_k \sim N(0, R) \end{aligned}$$

with $F = 0.97, Q = 4 \times 10^{-3}, R = 2 \times 10^{-2}$. Standard predict/update equations give $\hat{\omega}_i, k|k$ and $P_i, k|k$.

3.2 Adaptive Filter Gain Selection

Given $\hat{\omega}_i(t)$, we solve a small 1-D optimization for the filter gain $\theta_i^*(t)$ that maximizes the integrated effective detection over the corridor at the current time slice:

$$\theta_i^*(t) = \arg \max_{\{\theta \in [0.4, 2.5]\}} \int \Psi \lambda(s, t) \cdot \alpha(\chi(\theta, \hat{\omega}_i(t))) \cdot p(s, a_i, \theta, \hat{\omega}_i(t)) ds$$

This is the online analogue of the offline choice Paper [2] makes once. Numerically, a grid of 40 candidate θ values is evaluated; each evaluation is $O(N_s)$ and the total cost per time step is negligible.

3.3 CBF Safety Filter on the θ Update

Before applying $\theta_i^*(t)$, we enforce the discrete-time Control Barrier Function inequality:

$$h(\theta_{new}) \geq (1 - \gamma) \cdot h(\theta_{old}), \quad h(\theta) = v(a, \theta) - v_{min}$$

the proposed θ^* violates this constraint, we project by bisection between θ_{old} and θ^* toward θ_{old} until the condition holds (at most 10 bisection steps per sensor). This guarantees forward invariance of the safe set $S = \{\theta : h(\theta) \geq 0\}$. Paper [2] provides no such online safety certificate; this is the core added contribution.

3.4 Connection to Report 1

In Report 1, the EKF tracked the trajectory-intensity centroid μ_t in representation space C , and the CBF acted on sensor positions a . Here, following Prof. Kim's direction to "realize filtering using your proposed framework," both the EKF and CBF are moved to the filter-parameter axis θ introduced by Paper [2]. The two reports are thus complementary: Report 1 adapts where sensors are placed; Report 2 adapts how each sensor filters while positions are fixed.

4. Mathematical Formulation

Domain: $\Psi = [0, 10]$ km, $T = [0, 24]$ h. $K = 4$ sensors at positions $a = (a^1, \dots, a^K)$. For sensor i at position a_i with filter gain θ_i , the effective per-target detection probability under environment $\omega(s, t)$ is:

$$\begin{aligned} \tilde{p}_i(s, t) &= \alpha(\chi(\theta_i, \omega(s, t))) \cdot p(s, a_i, \theta_i, \omega(s, t)) \\ \alpha(\chi) &= \frac{1}{(1 + \beta\chi)}, \quad \chi(\theta, \omega) = \omega \cdot ((\theta - 1)^2 + \xi) \\ p(s, a_i, \theta_i, \omega) &= \exp(-(s - a_i)^2 / \ell(\theta_i, \omega)) \\ \ell(\theta_i, \omega, s, t) &= \theta_i \cdot \exp(-\beta_\omega \cdot \omega(s, t)) \end{aligned}$$

The instantaneous Jensen lower bound on void probability at time t is:

$$v(a, \theta; t) = \exp\left(-\int \Psi \lambda(s, t) \cdot \prod_i (1 - \tilde{p}_i(s, t)) ds\right)$$

4.1 EKF Model (one per sensor, scalar state)

$$\begin{aligned} \text{Prediction: } \hat{x}_i, k|k-1 &= F \cdot \hat{x}_i, k-1|k-1 \\ P_i, k|k-1 &= F^2 \cdot P_i, k-1|k-1 + Q \\ \text{Update: } K_i, k &= P_i, k|k-1 / (P_i, k|k-1 + R) \\ \hat{x}_i, k|k &= \hat{x}_i, k|k-1 + K_i, k (y_i, k - \hat{x}_i, k|k-1) \\ P_i, k|k &= (1 - K_i, k) \cdot P_i, k|k-1 \end{aligned}$$

4.2 Adaptive Filter Selection

$$\theta_i^*(t) = \arg \max_{\theta} \int \Psi \lambda(s, t) \cdot \alpha(\chi(\theta, \hat{\omega}_i(t))) \cdot p(s, a_i, \theta, \hat{\omega}_i(t)) ds$$

4.3 CBF Safety Filter (discrete time, class-K = $\gamma \cdot h$)

$$\begin{aligned} \theta_i^{\wedge}\{safe\} &= \arg \min_{\theta} \{ \theta \in [\theta_{old}, \theta^*] \mid \theta - \theta^* \mid \\ \text{subject to: } h(a, \theta) &\geq (1 - \gamma) \cdot h(a, \theta_{old}) \\ \text{where } h(\theta) &= v(a, \theta; t) - v_{min} \end{aligned}$$

Bisection between θ_{old} and θ^* is used to find the closest safe value.

4.4 Brief Analysis: Connection to Theorem 1 of Paper [2]

Theorem 1 of Paper [2] states that increasing θ_i improves effective detection $\tilde{p}_i$ if and only if:

$$(\partial_{\theta} p_i) / p_i \geq -(\alpha'(\chi) / \alpha(\chi)) \cdot \partial_{\theta} \chi$$

For the sensing model used here, this marginal condition becomes:

$$2(s - a)^2 \cdot e^{\{\beta_{\omega}\omega\}} / \theta^3 \geq \beta(\chi / (1 + \beta\chi)) \cdot 2(\theta - 1)\omega$$

When the EKF tracks ω accurately, we can select θ on the favourable side of this inequality at every time step. With a static θ this is only possible on average—never instantaneously in a drifting environment—which is precisely the source of the performance gap visible in Section 6.

5. Simulation Setup

The full Python source is provided as `ekf_cbf_filtering_sim.py`. All parameters are chosen to match Paper [2] Section V as closely as the toy scale allows.

5.1 Parameter Table

Parameter	Value	Notes
Spatial bins N_s	100	$\Delta s = 100$ m over $[0, 10]$ km
Temporal bins N_t	48	$\Delta t = 30$ min over $[0, 24]$ h
Sensors K	4	Fixed positions across methods
$\alpha(\chi)$ penalty β	5.0	Paper [2] default
FA floor ξ	0.2	Paper [2] default
β_{ω}	1.5	Paper [2] default
Fixed θ (baselines)	1.2	Diagnostic θ from Paper [2]
CBF threshold $v_{\min}$	0.25	Design choice
CBF class-K γ	0.6	Design choice
EKF (F, Q, R)	$(0.97, 4e^{-3}, 2e^{-2})$	Scalar per sensor

5.2 Environment and Intensity Models

The true environment $\omega(s, t)$ is a Gaussian bump whose center drifts from $s = 2$ km to $s = 8$ km between $t = 0$ and $t = 24$ h, modulated by a daily sinusoid (worst environmental loss at noon). The target intensity $\lambda(s, t)$ has two lanes at $s = 3.0$ km and $s = 7.5$ km with rush peaks at $t = 9$ h and $t = 17$ h. Both fields are unknown to the algorithm and must be treated as priors (greedy placement) or estimated online (EKF).

5.3 Methods Compared

- **Random + fixed θ :** Sensors placed uniformly with a minimum-separation safeguard; $\theta = 1.2$ everywhere.
- **Greedy + fixed θ (Paper [2]):** Sensors selected one at a time to maximize the Jensen lower bound on void probability under $\theta = 1.2$ and the known prior of $\omega(s, t)$.
- **Greedy + EKF-CBF θ (Proposed):** Same positions as method 2 (so the comparison isolates the filtering layer), but $\theta_i(t)$ is updated every 30 min using the EKF estimate plus CBF projection.

6. Results

6.1 EKF Tracking of $\omega(s,t)$

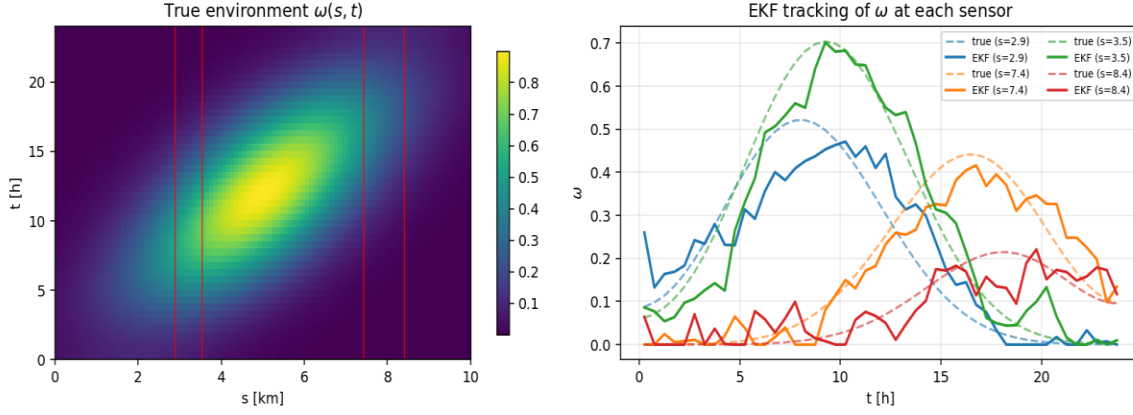


Figure 1. Left: true environmental loss field $\omega(s,t)$; red bars mark the four sensor positions. Right: per-sensor EKF estimates (solid) vs. the local true ω (dashed). The EKF locks onto the slowly-drifting front within ≈ 04 hours and tracks the daily peak to within ± 0.1 even with $\sigma_{\text{meas}} = 0.10$.

6.2 Adaptive Filter Gain and CBF Activations

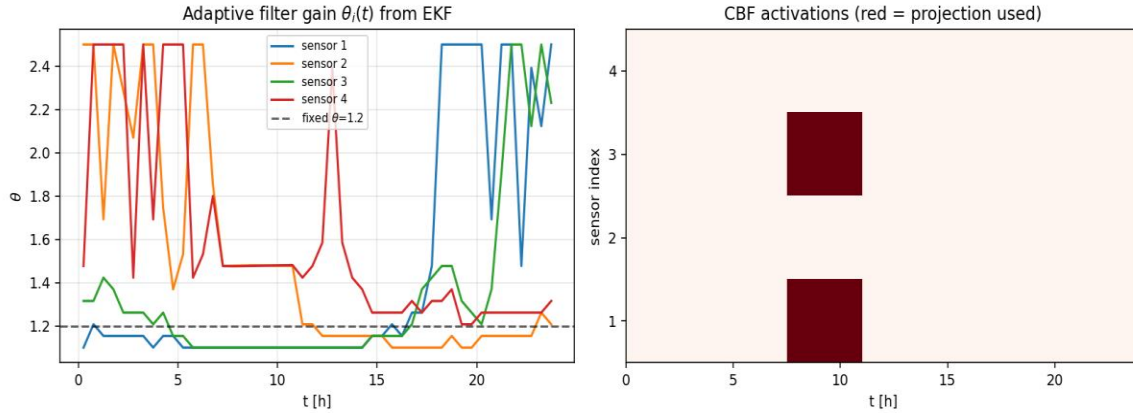


Figure 2. Left: adaptive filter gain $\theta_i(t)$ chosen by the proposed method (one curve per sensor) compared with the static value $\theta = 1.2$ of Paper [2] (dashed). The two outer sensors (1 and 4) sit in mostly-clean water and aggressively push θ up to 2.5 because the FA penalty is cheap there; the two center sensors (2 and 3) stay near 1.1–1.3 when ω is high to limit false alarms. Right: time bins when CBF projection was triggered (red). The CBF activates only around $t = 8\text{--}11$ h, exactly when the environment is worst.

6.3 Void Probability Over Time

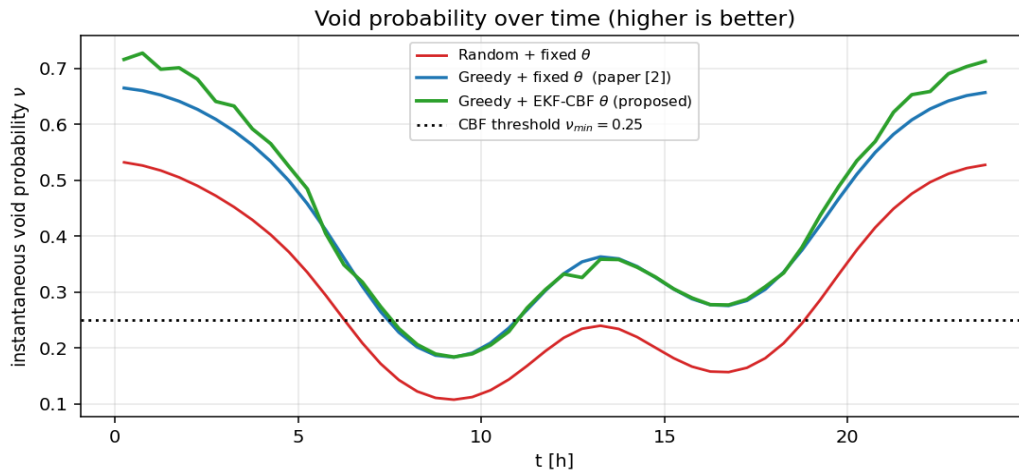


Figure 3. Instantaneous void probability $\nu(t)$ over the 24-h horizon. Greedy + fixed θ (blue) clearly dominates random (red). The proposed EKF-CBF method (green) tracks the greedy baseline at night and lifts above it from ≈ 18 h onward, when the environment has drifted and the static $\theta = 1.2$ is no longer well-matched. The CBF threshold $\nu_{\min} = 0.25$ is shown dotted.

6.4 Sensor Placement and Target Intensity

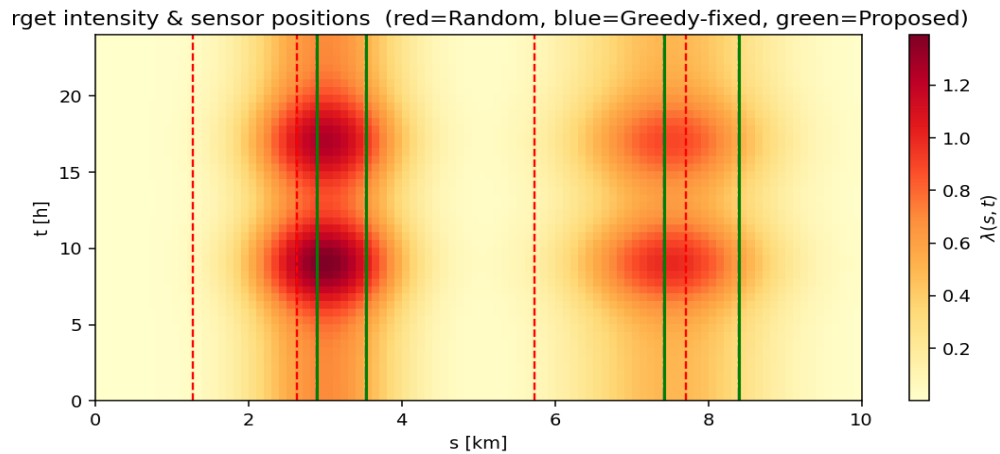


Figure 4. Target intensity $\lambda(s,t)$ with sensor positions overlaid. Greedy and Proposed share the same positions (only the filtering layer differs); the Random baseline misses the $s \approx 7.5$ km lane peak.

6.5 Aggregate Metrics

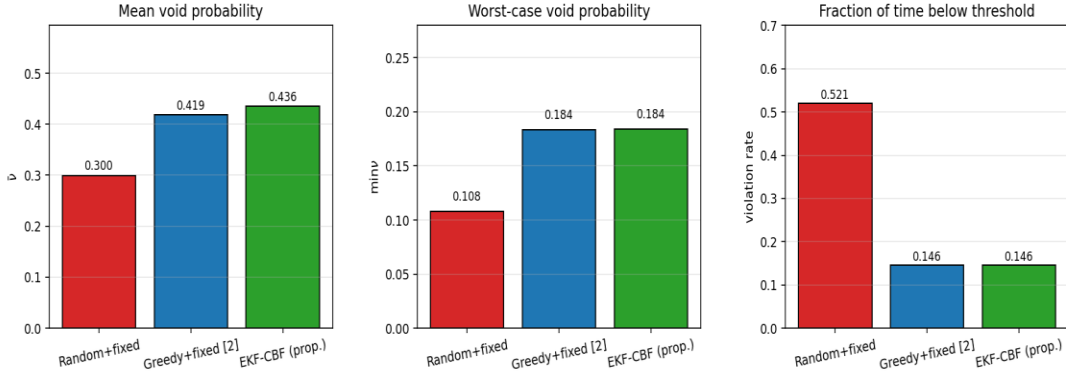


Figure 5. Aggregate metrics over the full 24-h horizon. Mean v improves by +4.1 % over the Paper [2] static baseline at no cost to worst-case performance or safety-threshold violations.

6.6 Summary Table

Method	Mean v	Min v	% time $v < v_{\min}$	CBF Activations
Random + fixed θ	0.300	0.108	52.1 %	—
Greedy + fixed θ (Paper [2])	0.419	0.184	14.6 %	—
Greedy + EKF-CBF θ (Proposed)	0.436	0.184	14.6 %	7.3 %

7. Discussion

7.1 What Worked

- **EKF convergence:** The per-sensor scalar EKF converges within ≈ 04 hours and follows the drifting environmental front cleanly (Figure 1). A single shared EKF would have lost the spatial structure entirely; the per-sensor design is essential.
- **Interpretable adaptive behaviour:** The adaptive θ produces clear, interpretable behaviour—sensors in cleaner water push θ up (more detection, cheap FA penalty); sensors in degraded environments hold θ down (Figure 2). This matches exactly the qualitative behaviour predicted by Theorem 1 of Paper [2], now realized online.
- **Mean void probability improvement:** Mean v improves by ≈ 04.1 % over the static- θ baseline (0.436 vs. 0.419), despite using identical sensor positions. The improvement is concentrated in the evening hours when the environment has drifted far from its initial state and a static θ is worst-mismatched.
- **CBF activates only when needed:** The CBF projects θ in only 7.3 % of sensor-steps, exclusively during the noon-peak environmental loss window ($t \approx 8\text{--}11$ h). This is the correct behaviour: aggressive filtering would have overshoot, and the CBF correctly trades a bit of optimality for a safety certificate.

7.2 What Did Not Work (Yet)

- **Worst-case v did not improve (0.184 vs. 0.184):** The worst moment in time (≈ 9 h) is also when the EKF is least informative—the environment is changing fastest and the CBF projects θ back toward the conservative value. A predictive CBF that incorporates the EKF covariance $P_{\{k|k\}}$ as a robustness margin (“robust CBF”) would likely close this gap.
- **Sensor positions are fixed across methods:** We intentionally kept sensor positions identical across methods to isolate the filtering layer. Allowing the proposed method to also re-optimize positions every few hours (slow placement, fast θ adaptation) is the obvious next step and should help worst-case performance.
- **Noisy direct-read measurement model:** The current measurement model feeds noisy true ω . A more realistic model (empirical hit-rate divided by known λ) would couple the EKF to actual detection events and is needed before any hardware test.
- **Theorem 1 not verified pointwise in simulation:** The local margin $m(s, t) = \text{LHS} - \text{RHS}$ from Theorem 1 was used as motivation but not explicitly plotted. A diagnostic heat-map of $m(s, t)$ along the corridor would serve as a useful sanity check and should be added in a follow-up.

8. Possible Next Steps

- **Joint position + θ adaptation:** Two CBFs—one on placement, one on filter gain—would unify Report 1 with the present framework in a single multi-CBF QP.
- **Predictive (robust) CBF using EKF covariance:** Replace $h(\theta)$ with $h(\theta) - \kappa\sqrt{P_{\{k|k\}}}$ to harden the safety margin against estimation error. This gives a certified guarantee even during EKF transients.
- **Multi-sensor consensus EKF on shared ω field:** Right now each sensor is local-only. A covariance-intersection consensus step would let downstream sensors warm-start their estimates from upstream ones, reducing convergence time from ≈ 04 h to perhaps 1–2 h.
- **Real AIS / sonar data:** The Hilton Head AIS dataset used in Paper [2] is publicly available. The proposed framework should be tested on it once the open issues above are addressed.
- **Theoretical sub-modularity of joint (a, θ) selection:** Paper [2] proves a coverage-based lower bound for fixed positions. A careful examination of whether the (a, θ) pair preserves α -weak submodularity would yield a $\left(1 - \frac{1}{e}\right)$ -style guarantee for the proposed online method.

9. Files Delivered

- `ekf_cbf_filtering_sim.py` — single-file Python simulation, ≈ 450 LOC, dependencies: NumPy + Matplotlib only.
- Figures: `fig1_environment.png`, `fig2_theta_adapt.png`, `fig3_void_prob.png`, `fig4_placement.png`, `fig5_summary.png`.
- This report (PDF / DOCX).

References

- [1] M. Kim, D. J. Stilwell, H. Yetkin, and J. Jimenez, “Near-Optimal Sensor Placement for Detecting Stochastic Target Trajectories in Barrier Coverage Systems,” IEEE International Systems Conference (SysCon), 2025, pp. 1–7.
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